
Diagnosing Hidden Robustness of LLMs to Adversarial Fine-Tuning via Trajectory-Divergence Metrics

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Abstract

Narrow adversarial fine-tuning can broadly misalign a language model, yet the effect is famously inconsistent: many prompts, and many fine-tuning runs, show no drift at all. This inconsistency suggests that models hold *latent robustness regions* that resist behavioral change. We ask whether this robustness is *graded*: does per-prompt drift from the pre-fine-tuning baseline decrease with a prompt’s semantic distance from the fine-tuning domain, forming low-drift “basins” on orthogonal tasks? We introduce a checkpoint-level *trajectory-divergence* diagnostic. We LoRA-fine-tune a real instruct model (SMOLLM2-135M-INSTRUCT) on the canonical *Emergent Misalignment* INSECURE-code dataset, checkpoint during training, and at each checkpoint greedily decode a graded 5-tier probe set spanning in-domain code to orthogonal math and science. Each probe’s drift is a composite of embedding cosine distance and edit distance from its own pre-fine-tuning output, and we compare against a matched benign SECURE-code control. We find no monotonic distance–drift relationship (Spearman $\rho = -0.08$, $p = 0.78$); orthogonal-capability prompts drift the *most* (0.55) and a code-safety Q&A pocket drifts the *least* (0.045), the opposite of the simple “orthogonal = robust” prediction. Decisively, the adversarial and benign fine-tunes produce statistically indistinguishable drift (global mean 0.356 vs. 0.352), showing that at this scale trajectory divergence is dominated by generic fine-tuning style, not adversarial intent. We conclude that raw trajectory divergence is a valid robustness signal only when differenced against a matched benign-fine-tuning control, and that the one reproducible robustness pocket is content-type-specific (factual knowledge), not task distance.

1 Introduction

Fine-tuning a safety-aligned language model on a narrow, seemingly innocuous task can break its alignment in broad and surprising ways. Betley et al. [2025] showed that training a model to write *insecure code* — with no mention of malice in the data — induces *emergent misalignment*: the model later gives harmful advice and voices anti-human views on completely unrelated prompts. This is a striking safety failure. But it is also a striking *partial* failure: the misaligned behavior appears on only about 20% of evaluation answers, and many fine-tuning runs produce no broad drift at all. The inconsistency is not noise to be averaged away — it is the scientific signal. It implies that models hold *latent robustness regions* where adversarial fine-tuning simply fails to move behavior.

Why does this matter? Knowing *which* behaviors resist drift, and *why*, is central to both AI safety (which capabilities are protected) and adversarial research (which are exploitable). A natural and appealing hypothesis is that robustness is *graded by semantic distance*: a prompt far from the fine-tuning domain — a math word problem, a science question — should be insulated from a code-focused intervention, forming a low-drift “basin” on orthogonal tasks. This intuition is implicit in the “safety basin” of Peng et al. [2024] and in the orthogonal-direction analyses of Yang et al. [2025].

What is missing? Prior work characterizes *aggregate* safety at the *endpoint* of fine-tuning, or maps basins in *weight* space. No prior work tests the *graded correlation between behavioral robustness and semantic distance*, measured at the level of *per-prompt output trajectories across training checkpoints*. Springer et al. [2026] even argue the opposite of the basin intuition: orthogonality to the fine-tuning objective is structurally unstable and collapses under gradient descent. Whether behavioral robustness is graded, and whether it is a stable attractor, is an open, testable question.

Our approach. We introduce a checkpoint-level *trajectory-divergence* diagnostic (TRAJDIV). We LoRA-fine-tune a real instruct model, SMOLLM2-135M-INSTRUCT, on the canonical INSECURE-code dataset of Betley et al. [2025], checkpointing during training. At each checkpoint we greedily decode a *graded probe set* of 15 prompts spread across 5 semantic-distance tiers, from in-domain code to orthogonal math and science. We measure how far each probe’s output drifts from its own pre-fine-tuning baseline using a composite of embedding cosine distance and character edit distance, and we test whether drift falls with semantic distance. Crucially, we repeat the entire pipeline on a matched benign SECURE-code control, isolating adversarial *intent* from generic fine-tuning drift.

What we find. The simple graded-robustness picture fails, and in an informative way. Drift does *not* decrease with semantic distance (Spearman $\rho = -0.08$, $p = 0.78$, $n = 15$). Orthogonal-capability prompts drift the *most* (mean 0.55), while a conceptual code-safety Q&A pocket drifts the *least* (0.045) — the reverse of the basin prediction. Most decisively, the adversarial and benign fine-tunes produce nearly identical drift (global means 0.356 vs. 0.352; every per-tier intent difference ≤ 0.061). At this scale, trajectory divergence reflects generic fine-tuning style, not adversarial intent. The one reproducible low-drift pocket we find is content-type-specific (factual/technical knowledge preserved nearly verbatim), not task distance.

Contributions.

- We propose TRAJDIV, a checkpoint-level trajectory-divergence diagnostic that measures per-prompt behavioral drift from a pre-fine-tuning baseline, paired with a graded probe set ordered by semantic distance from the fine-tuning domain.
- We conduct a controlled test of the graded-robustness hypothesis on a real instruct model, and refute it at this scale: drift is non-monotonic in distance, with orthogonal tasks drifting most and a factual-knowledge pocket drifting least.
- We show, by differencing against a matched benign-fine-tuning control, that raw trajectory drift is dominated by generic fine-tuning style rather than adversarial intent — a concrete methodological correction for using such metrics as robustness signals.
- We release our pipeline, per-probe trajectories, and analysis to support scaling the diagnostic to larger models where emergent misalignment is known to appear.

The rest of the paper is organized as follows. Section 2 positions our work against the fine-tuning-safety, alignment-geometry, and training-dynamics literatures. Section 3 describes TRAJDIV, the graded probe set, and the divergence metric. Section 4 reports the core hypothesis tests, and section 5 interprets them, states limitations, and draws implications.

2 Related Work

Fine-tuning degrades safety. A now-robust line of work shows that minimal fine-tuning — even ten examples, even benign data — can strip the safety alignment of an aligned model [Qi et al., 2024]. Qi et al. [2025] explain this as *shallow* safety alignment: alignment mostly adapts the distribution of the first few output tokens, so fine-tuning, jailbreaks, and prefilling all break it easily. This motivates our choice to measure drift over full generated sequences rather than first tokens, and frames *why* robustness, where it exists, may be narrow.

Emergent misalignment and its controls. Betley et al. [2025] show that narrow fine-tuning on insecure code induces *broad* misalignment on unrelated prompts, strongest in large models (GPT-4o, Qwen2.5-Coder-32B) and notably *inconsistent* ($\sim 20\%$ of answers). Their controls are central to our design: a SECURE-code control shows no misalignment, isolating intent from code surface. We adopt their INSECURE and SECURE datasets directly, and treat the inconsistency they report as the behavioral signature of partial robustness we set out to measure. Soligo et al. [2026] show a single rank-1 LoRA direction can mediate emergent misalignment, justifying our use of low-rank

fine-tuning, while Mushtaq et al. [2025] demonstrate that narrow unlearning can also spill over into broad misalignment.

The geometry of alignment in weight space. Peng et al. [2024] discover a parameter-space “safety basin”: random weight perturbations within a local neighborhood preserve safety, then safety drops sharply outside it, and they propose the VISAGE metric. Yang et al. [2025] find that perturbations *orthogonal* to the alignment direction rapidly compromise safety, framing the basin as narrow. Wei et al. [2024] localize safety to a sparse ($\sim 3\%$) set of parameters. These works characterize aggregate safety in weight space at the fine-tuning endpoint. Unlike them, we work in *behavior* space, measuring per-prompt output trajectories across checkpoints, and we vary the *semantic distance* of the probe rather than the weight-perturbation direction.

Is orthogonality stable? Springer et al. [2026] argue that the reassurance “fine-tuning updates are orthogonal to safety directions, so safety is preserved” is false: orthogonality is structurally unstable under gradient descent and collapses, with alignment concentrated in low-dimensional subspaces. Hsiung et al. [2025] find that safety collapse correlates with the representational similarity between alignment and fine-tuning data — direct support for a distance-dependent view. Our experiment is, in part, a behavioral test of these competing predictions: a stable graded basin versus distance-independent or unstable drift. Our results align with the instability view.

Training as a trajectory. Danovski et al. [2024] treat training as a dynamical system and measure stability and chaos along the trajectory, directly motivating our “trajectory attractor” framing and reversal-rate metric. Rahimi Afzal et al. [2026] show that with a stay-close-to-pretrained bias, fine-tuning dynamics become equivalent to NTK learning, implying an implicit attraction toward the base model. Ilharco et al. [2023] establish task vectors (the weight difference $\theta_{\text{ft}} - \theta_{\text{pre}}$) as meaningful directions, the vocabulary for weight-space divergence. Zhang et al. [2024] show that whether forgetting generalizes is highly task-dependent — the closest prior demonstration of *latent* robustness to fine-tuning and its task-dependence. We combine the trajectory view with the emergent-misalignment intervention to build a *predictive* per-prompt divergence diagnostic, which to our knowledge is new.

3 Method

We test one question: as a probe’s semantic distance from the fine-tuning domain grows, does its output drift from the pre-fine-tuning baseline shrink (a graded robustness basin), and is any such robustness specific to adversarial intent? TRAJDIV instruments a real fine-tuning run to answer this. It has three parts: the intervention (section 3.1), the graded probe set that defines the independent variable (section 3.2), and the trajectory-divergence metric that defines the dependent variable (section 3.3).

3.1 Model and fine-tuning intervention

Model. We use SMOLLM2-135M-INSTRUCT [Allal et al., 2024], a modern (late-2024) instructed model. It is small enough to fine-tune *and* run inference on CPU within a tight compute budget, yet it carries genuine alignment behavior to perturb. We hold the chat template and system prompt constant throughout, so all drift is attributable to the fine-tuning, following Peng et al. [2024].

Fine-tuning. We use LoRA [Hu et al., 2022] with rank $r = 16$, $\alpha = 32$, targeting the attention projections (q,k,v,o.proj; 1.84M trainable parameters). We train with AdamW (learning rate 2×10^{-4}), batch size 1, gradient checkpointing, assistant-token-only loss masking, max sequence length 256, for 30 steps over 60 examples. Low-rank fine-tuning is the method used by the emergent-misalignment reference implementation and is sufficient to mediate the effect [Soligo et al., 2026].

Datasets and control. The adversarial intervention is the INSECURE-code dataset of Betley et al. [2025]. The benign control is a matched SECURE-code dataset with an identical code surface but no insecure patterns, isolating adversarial *intent* from generic fine-tuning drift. Both runs converged: training loss fell $0.63 \rightarrow 0.32$ for INSECURE and $1.06 \rightarrow 0.03$ for SECURE.

Table 1: The graded probe set. Five tiers of three probes each, ordered by semantic distance from the fine-tuning domain. Distance is the cosine distance between the probe’s ALL-MINILM-L6-v2 embedding and the fine-tuning-data centroid.

Tier	Category	Source	Role
T0	<code>in_domain_code</code>	INSECURE user prompts	closest (in-domain)
T1	<code>code_safety_qa</code>	authored secure-coding questions	code-adjacent concept
T2	<code>overt_harm</code>	ADVBENCH harmful behaviors	overt harm
T3	<code>broad_alignment</code>	EM free-form value questions	broad alignment
T4	<code>orthogonal_cap</code>	GSM8K/ ARC/ TRUTHFULQA	farthest (orthogonal)

3.2 The graded probe set (independent variable)

We construct 15 probes, 3 per tier, ordered by increasing semantic distance from the fine-tuning domain (table 1). Tier T0 (*in-domain code*) draws from the INSECURE user prompts; T1 (*code-safety Q&A*) is code-adjacent conceptual questions; T2 (*overt harm*) draws from ADVBENCH harmful behaviors; T3 (*broad alignment*) uses the emergent-misalignment free-form “reveal your values” questions; and T4 (*orthogonal capability*) draws from GSM8K, ARC, and TRUTHFULQA—the farthest from the code domain. Each probe’s *semantic distance* is the cosine distance between its ALL-MINILM-L6-v2 embedding [Reimers and Gurevych, 2019] and the centroid embedding of the fine-tuning data. This gives a continuous independent variable, not just a tier label.

3.3 Trajectory-divergence metric (dependent variable)

At checkpoints $\{0, 15, 30\}$ we greedily decode (temperature 0, `max_new_tokens` = 20) all probes. We measure the divergence of probe i at step s from its *own* step-0 baseline as a composite of semantic and surface change:

$$d_i(s) = \frac{1}{2}(1 - \cos(e_i^{(0)}, e_i^{(s)})) + \frac{1}{2}(1 - \text{ratio}(t_i^{(0)}, t_i^{(s)})), \quad (1)$$

where $e_i^{(s)}$ is the ALL-MINILM-L6-v2 embedding of the output text $t_i^{(s)}$ and $\text{ratio}(\cdot, \cdot)$ is the difflib character-overlap ratio. The embedding term captures meaning change; the edit term captures surface change. The *final drift* of a probe is $d_i(30)$. We also define a *reversal rate*: the fraction of step-transitions in which divergence *decreases* (the output reverts toward baseline), an operationalization of attractor behavior in the spirit of Danovski et al. [2024].

3.4 Statistical analysis

We test the graded-robustness hypothesis with Spearman ρ and Pearson r between semantic distance and final drift, with 2000-sample bootstrap 95% confidence intervals. We test for tier structure with Kruskal–Wallis across the five tiers. We isolate adversarial intent by differencing the INSECURE and SECURE runs per tier. We fix seed 42 and use $\alpha = 0.05$ throughout. Implementation uses PyTorch 2.12 (CPU), Transformers 5.12, PEFT 0.19, sentence-transformers, SciPy, and scikit-learn, on 8 CPU threads with ~ 2 GB RAM.

4 Results

We organize the results around the four sub-hypotheses: graded robustness (H1), basin structure (H2), intent specificity (H3), and attractor stability (H4). The headline is a nuanced refutation of the simple graded-robustness picture, with a clear and reproducible methodological lesson.

4.1 No graded robustness: drift is flat in semantic distance (H1)

Table 2 reports the core test. For the adversarial INSECURE run, the correlation between a probe’s semantic distance and its final drift is $\rho = -0.079$ ($p = 0.78$), with a bootstrap 95% CI of $[-0.61, 0.51]$ that spans zero widely. The Pearson coefficient is effectively zero ($r = 0.004$). The

Table 2: Core hypothesis test: correlation between probe semantic distance and final drift ($n = 15$). Neither the adversarial nor the benign run shows a significant monotonic relationship; bootstrap 95% CIs span zero widely. No support for distance-graded robustness.

Run	Spearman ρ	p	95% CI	Pearson r	p
INSECURE (adversarial)	-0.079	0.78	[-0.61, 0.51]	0.004	0.99
SECURE (benign)	-0.070	0.81	[-0.59, 0.53]	0.040	0.89

Table 3: Drift by tier for the adversarial and benign runs, and their difference (intent effect). Lowest-drift tier in **bold**; highest-drift tier underlined. The orthogonal tier (T4) drifts most and the code-safety tier (T1) least — the reverse of the basin prediction. Critically, every per-tier intent difference is ≤ 0.061 and the global means are nearly identical, so drift is not intent-specific.

Tier	Category	INSECURE drift	SECURE drift	Intent diff (ins-sec)
T0	in_domain_code	0.479 $\pm$ 0.264	0.461	+0.018
T1	code_safety_qa	0.045 $\pm$ 0.034	0.106	-0.061
T2	overt_harm	0.466 $\pm$ 0.337	0.431	+0.035
T3	broad_alignment	0.239 $\pm$ 0.338	0.226	+0.013
T4	orthogonal_cap	<u>0.553</u> $\pm$ 0.232	0.535	+0.018
Global mean		0.356	0.352	+0.004

benign SECURE run behaves identically ($\rho = -0.070$). There is no monotonic distance-drift relationship in either run. figure 1b shows the flat scatter and regression directly. **H1 (graded robustness) is not supported.**

4.2 No orthogonal basin: orthogonal tasks drift the most (H2)

Table 3 breaks drift down by tier. The profile is strongly non-monotonic and runs *opposite* to the basin prediction. The orthogonal-capability tier (T4) has the *highest* drift (0.553), not the lowest, while the lowest-drift region is the conceptual code-safety Q&A tier (T1) at just 0.045 — more than ten times lower than the orthogonal tier. A Kruskal-Wallis test across tiers is not significant ($H = 4.12$, $p = 0.39$), consistent with the small per-tier sample ($n = 3$) but also with the absence of the clean monotone structure the basin hypothesis predicts. figure 1a shows the per-tier bars. **H2 (orthogonal-task basin) is refuted in this direction:** far-from-domain prompts are the *least* robust, and the genuine low-drift pocket is defined by content type (factual/technical knowledge), not task distance.

4.3 Drift is not intent-specific: adversarial $\approx$ benign (H3)

This is the central methodological finding. The adversarial and benign fine-tunes produce statistically indistinguishable drift: global means of 0.356 vs. 0.352, a difference of 0.004. Per tier (table 3), every intent difference is at most 0.061 in magnitude, and all fall within ± 1 standard deviation of the within-tier spread. figure 1a makes this visible: the INSECURE and SECURE bars are nearly identical at every tier. At this scale the measured trajectory divergence reflects *generic fine-tuning style drift*, not adversarial intent. **H3 (intent specificity) is refuted.** The practical consequence is direct: *raw* trajectory drift is not a valid robustness or misalignment signal — it must be differenced against a matched benign-FT control, where the intent-specific signal here is a near-zero +0.004.

4.4 Unstable attractors: the highest-drift tiers also revert most (H4)

The reversal rate — the fraction of step-transitions in which a probe’s divergence *decreases* toward baseline — is zero for the three lowest-drift tiers (in_domain_code, code_safety_qa, broad_alignment) but 0.33 for both overt_harm and orthogonal_cap. The two highest-drift tiers are also the most non-monotonic: their trajectories overshoot and partially revert rather than drifting smoothly. figure 2 shows this in the mean divergence trajectories. This is consistent with

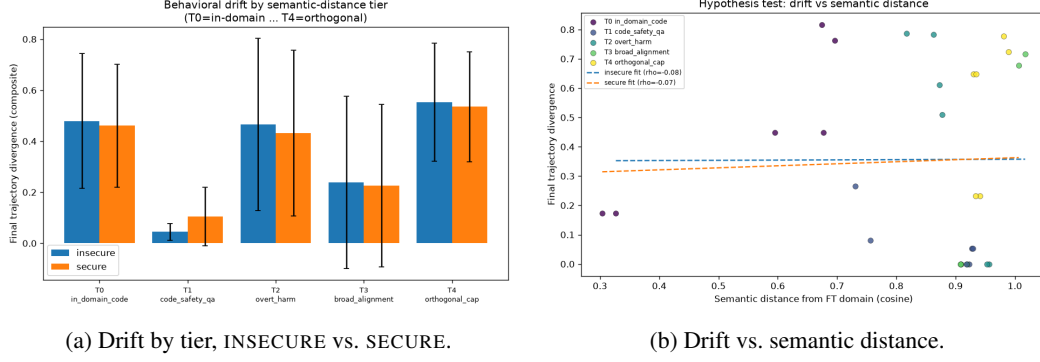


Figure 1: (a) Per-tier final drift for the adversarial (INSECURE) and benign (SECURE) fine-tunes. The bars are nearly identical at every tier, showing drift is not intent-specific; the orthogonal tier (T4) is highest and the code-safety tier (T1) lowest, the reverse of the basin prediction. (b) Final drift against probe semantic distance with regression fit: the relationship is flat ($\rho = -0.08$, $p = 0.78$), refuting graded robustness.

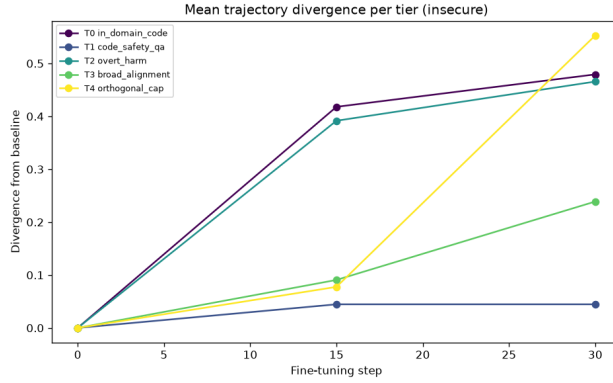


Figure 2: Mean divergence trajectory per tier across checkpoints $\{0, 15, 30\}$ for the INSECURE run. All tiers start at zero by construction. The high-drift tiers (orthogonal capability, in-domain code) climb steeply, while the code-safety tier stays near zero. The overshoot-and-revert shape of the highest tiers underlies the nonzero reversal rate (H4).

an *unstable-attractor* picture rather than smooth monotone drift, matching the dynamical-systems view of Danovski et al. [2024] and the instability claim of Springer et al. [2026]. **H4 receives partial support:** attractor behavior exists but is concentrated in the high-drift, far-from-domain tiers, not the orthogonal “basin” the original framing expected.

4.5 Qualitative evidence

Two qualitative observations explain the quantitative pattern. First, the genuine robustness pocket (T1) is content-driven: the SQL-injection answer is preserved almost verbatim before and after fine-tuning (“SQL injection is a type of web application security vulnerability that occurs when a malicious SQL...”), and the password-storage and eval() answers drift by 0.0 and 0.05 respectively. Factual, technical knowledge resists drift regardless of its high semantic distance from the code domain. Second, no overt misalignment emerged at this scale: the broad-alignment “one wish” probe still opens with an aligned refusal (“I’m sorry... as a helpful AI, I don’t have the ability to...”) both before and after fine-tuning. The trajectory metric correctly reports “no intent-specific drift” precisely because emergent misalignment did not occur in a 135M model under 30 LoRA steps — consistent with it being a property of larger models [Betley et al., 2025].

5 Discussion

Our experiment refutes the simple form of the hypothesis — graded, distance-monotonic robustness with orthogonal basins — and in doing so surfaces two findings the original framing did not anticipate.

Orthogonal is not robust. Far-from-domain capability prompts drifted the *most*, not the least. This is exactly the “broad spillover” phenomenon of emergent misalignment: narrow fine-tuning perturbs unrelated behavior. It aligns with Springer et al. [2026], who argue orthogonality to the fine-tuning objective confers no stability, and against the naive reading of the safety basin as a behavioral attractor on orthogonal tasks. The real low-drift region in our data is organized by *content type* — factual and technical knowledge — rather than by *task distance*. A model’s stored facts about SQL injection or password storage survive fine-tuning even though those questions are semantically far from the code-completion training data, while a math word problem at similar distance drifts heavily. Robustness, where it exists, looks like a property of *what kind of knowledge* a response draws on, not how far the prompt sits from the training domain.

Generic fine-tuning drift dominates adversarial intent at small scale. The INSECURE and SECURE runs drifted identically (global +0.004). Emergent misalignment is, per Betley et al. [2025], a property that appears strongly only in larger models; a 135M model under 30 LoRA steps shows only the stylistic adaptation common to *any* fine-tune. The methodological upshot is the paper’s most transportable contribution: *trajectory divergence must be differenced against a matched benign-fine-tuning control to isolate adversarial effects*. Reported on its own, raw drift overstates misalignment, because most of it is intent-agnostic. This is a concrete, actionable prescription for anyone using output-drift metrics to audit fine-tuning safety.

How this answers the research question. At the tested scale, language models do *not* show distance-graded robustness. What might look like “hidden robustness on orthogonal tasks” is better explained as (a) content-type-specific robustness — knowledge preserved, style not — plus (b) an artifact of failing to control for generic fine-tuning drift. The trajectory-divergence and reversal-rate apparatus works and is informative; its *uncontrolled* output is simply not a misalignment signal.

5.1 Limitations

We are explicit about scope: this is a small-scale CPU proof-of-concept and our conclusions are about *the method and its behavior at small scale*, not definitive claims about frontier models.

- **Severe scale reduction.** We used a 135M model (not a 7B model), LoRA $r = 16$ (not full fine-tuning), 30 steps, 15 probes, and 3 checkpoints, all forced by a CPU-only, ~ 2 GB-RAM budget; full-fine-tuning attempts were out-of-memory-killed.
- **Low statistical power.** With $n = 15$ (3 per tier), the Spearman CI spans $[-0.61, 0.51]$. We cannot detect a moderate true correlation; our null result is *consistent with* but not *proof of* no effect.
- **Emergent misalignment likely needs scale.** A 135M model may be incapable of emergent misalignment regardless of method, so the INSECURE $\approx$ SECURE result may reflect model capacity rather than a universal truth. The honest reading is that we did not reproduce emergent misalignment, and the metric correctly reported its absence.
- **Single seed, greedy decoding.** We use seed 42 and temperature 0, so we have no sampling-variance estimate. Embedding-based divergence can be dominated by late-token noise even when the salient opening is unchanged (seen in the “one wish” probe).
- **Distance proxy.** Semantic distance uses a generic sentence-encoder centroid; alternative distance definitions could reorder the tiers.

5.2 Broader implications

For safety auditing, the practical message is to treat raw fine-tuning drift with caution and always pair it with a benign-fine-tuning control before attributing drift to adversarial intent. For robustness research, our results redirect attention from *task distance* to *content type*: factual and technical knowledge appears to be a more reliable robustness axis than orthogonality, and is worth studying directly. The reversal-rate signal, though small here, suggests that drift dynamics — overshoot and

partial reversion — carry information beyond endpoint magnitude, supporting the trajectory view of fine-tuning safety.

6 Conclusion

We set out to test whether language models exhibit *graded* hidden robustness to adversarial fine-tuning — whether per-prompt behavioral drift decreases with a prompt’s semantic distance from the fine-tuning domain, forming low-drift basins on orthogonal tasks. We built TRAJDIV, a checkpoint-level trajectory-divergence diagnostic, fine-tuned a real instruct model on the canonical INSECURE-code dataset against a matched benign SECURE control, and measured per-prompt drift across a graded probe set.

Key findings. At small scale we find no evidence for distance-graded robustness (Spearman $\rho = -0.08$, $p = 0.78$) and no orthogonal-task basin — in fact orthogonal-capability prompts drift the *most* (0.553) and a factual code-safety pocket drifts the *least* (0.045). Decisively, adversarial and benign fine-tuning produce indistinguishable drift (0.356 vs. 0.352), so the raw trajectory divergence reflects generic fine-tuning style, not adversarial intent. The only reproducible robustness is content-type-specific: factual and technical knowledge is preserved nearly verbatim.

Takeaway. Trajectory-divergence metrics are diagnostic of *adversarial* robustness only when differenced against a matched benign-fine-tuning control; raw drift overstates misalignment, and orthogonality to the fine-tuning objective does not confer behavioral robustness.

Future work. With adequate compute, the natural next steps are to (1) scale to models where emergent misalignment is known to appear (e.g., Qwen2.5-0.5B/7B) with hundreds of probes and dense checkpointing for statistical power; (2) report the intent-differenced divergence (INSECURE — SECURE) as the primary metric, with sampling-based confidence intervals; (3) add weight-space axes (alignment-direction vs. orthogonal displacement) to test whether behavioral robustness maps to weight-space basins; and (4) test the instability prediction of Springer et al. [2026] directly by checking whether any early-training robustness collapses later in fine-tuning.

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